

Formulario de Ecuaciones Diferenciales e Integrales

Análisis Numérico / Ecuaciones Diferenciales

1. Variables separables

$$\frac{dy}{dx} = f(x)g(y)$$

Separar:

$$\frac{dy}{g(y)} = f(x) \, dx$$

Integrar ambos lados.

2. Homogéneas

Si

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right)$$

Cambio

$$y = ux$$

$$y' = u + x \frac{du}{dx}$$

Si aparece

$$F\left(\frac{x}{y}\right)$$

usar

$$x = vy$$

$$dx = v \, dy + y \, dv$$

3. Exactas

Forma

$$Mdx + Ndy = 0$$

Condición

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Solución

$$F(x, y) = C$$

donde

$$F_x = M, \quad F_y = N$$

4. Factor integrante

Si

$$M_y \neq N_x$$

Caso 1

$$\frac{N_x - M_y}{M} = f(y)$$

$$\mu(y) = e^{\int f(y) \, dy}$$

Caso 2

$$\frac{M_y - N_x}{N} = f(x)$$

$$\mu(x) = e^{\int f(x) \, dx}$$

5. Lineales

Forma

$$y' + P(x)y = Q(x)$$

Factor integrante

$$\mu(x) = e^{\int P(x) \, dx}$$

Después

$$(\mu y)' = \mu Q$$

6. Integración por partes

$$\int u \, dv = uv - \int v \, du$$

Regla LIATE

$$L > I > A > T > E$$

- Logarítmica
- Inversa
- Algebraica
- Trigonométrica
- Exponencial

7. Sustitución

Si aparece

$$f(g(x))g'(x)$$

hacer

$$u = g(x)$$

$$du = g'(x) dx$$

8. Fracciones parciales

Lineales

$$\frac{P(x)}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$$

Lineales repetidas

$$\frac{A}{x-a} + \frac{B}{(x-a)^2} + \cdots$$

Cuadrática irreducible

$$\frac{Ax+B}{x^2+bx+c}$$

9. Sustitución trigonométrica

$$\sqrt{a^2 - x^2}$$

$$x = a \sin \theta$$

$$\sqrt{a^2 + x^2}$$

$$x = a \tan \theta$$

$$\sqrt{x^2 - a^2}$$

$$x = a \sec \theta$$

10. Integrales básicas

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\int e^x dx = e^x + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C$$

11. Integrales trigonométricas

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \tan x dx = -\ln |\cos x| + C$$

$$\int \cot x dx = \ln |\sin x| + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \csc^2 x dx = -\cot x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \csc x \cot x dx = -\csc x + C$$

12. Hiperbólicas

$$\int \sinh x dx = \cosh x + C$$

$$\int \cosh x dx = \sinh x + C$$

13. Logaritmos

$$\ln(ab) = \ln a + \ln b$$

$$\ln\left(\frac{a}{b}\right) = \ln a - \ln b$$

$$\ln(a^n) = n \ln a$$

14. Exponenciales

$$e^{\ln x} = x$$

$$\ln(e^x) = x$$

$$e^{a+b} = e^a e^b$$

15. Derivadas importantes

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

16. Euler

$$y_{n+1} = y_n + h f(x_n, y_n)$$

17. Taylor

$$y(x+h) = y + hy' + \frac{h^2}{2}y'' + \frac{h^3}{6}y''' + \dots$$

18. Runge-Kutta 2

$$k_1 = h f(x, y)$$

$$k_2 = h f(x+h, y+k_1)$$

$$y_{n+1} = y + \frac{k_1 + k_2}{2}$$

19. Runge-Kutta 4

$$k_1 = h f(x, y)$$

$$k_2 = h f\left(x + \frac{h}{2}, y + \frac{k_1}{2}\right)$$

$$k_3 = h f\left(x + \frac{h}{2}, y + \frac{k_2}{2}\right)$$

$$k_4 = h f(x+h, y+k_3)$$

$$y_{n+1} = y + \frac{k_1 + 2k_2 + 2k_3 + k_4}{6}$$